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U SPArC Style Assignment

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1 Sentence that violates 'Understandable information load':

It was also demonstrated significant area savings and enhanced routing efficiency were achieved in both high-performance and high-density scenarios, while models indicate more than a four times reduction in IR-drop compared to conventional PDN.

Improved version of the sentence:

Studies also demonstrated significant area savings and enhanced routing efficiency in both high-performance and high-density scenarios. Furthermore, models indicate more than a four-fold reduction in IR-drop compared to conventional PDNs.

2 Sentence that violates 'Strong order':

As a result of the scaling of interconnects, an issue arises from the increased resistance.

Improved version of the sentence:

Interconnect scaling increases wire resistance, which becomes a growing concern at advanced nodes.

3 Sentence that violates 'Precise':

To address such limitations buried power rails (BPR) have been proposed as an early innovation.

Improved version of the sentence:

To address the growing resistance and routing congestion of front-side PDNs, buried power rails (BPR) have been proposed as an early innovation.

4 Sentence that violates 'Appropriate register':

This architecture has been used for multiple generations of integrated circuits, however continued scaling below 5 nm has revealed several limitations that have made reliable power delivery more difficult.

Improved version of the sentence:

This architecture has served multiple generations of integrated circuits. However, continued scaling below 5 nm has revealed several limitations that make reliable power delivery increasingly challenging.

5 Sentence that violates 'Concise':

Another key limitation that arises due to scaling is the increased competition for the limited routing resources available for power and signal interconnects.

Improved version of the sentence:

Scaling also intensifies competition between power and signal interconnects for routing resources.